

A NOTE ON A toy model of unemployment

Content of the Handout

1	Overview	p. 1
2	The model	p. 2

1. Overview

The existence of involuntary unemployment has long been recognized as one the main ills of modern industrialized economies, making old Keynesian economics relevant in analyzing and fixing unemployment untill the 70s. [Keynes \(1937\)](#) argued that unemployment was the result of weak aggregate demand: radical uncertainty about future states of the economy leads entrepreneurs and households to hold precautionary saving, making aggregate consumption and investment too weak for the economy to reach full employment.

At the end of the 60s, [Phelps \(1967\)](#), Milton Friedman (among others) questioned way the Phillips curve was designed, and criticized particularly the absence of expectations in the formation process of prices and inflation. Oil price shocks in the 70s proved how wrong were some assumptions of the keynesian model, in particular the price rigidity and the absence of expectations. [Lucas \(1976\)](#) contributed to the building a new framework where involuntary unemployment was absent. Despite the central role of unemployment in the policy debate, that variable has been absent from this new generation of models that have become the workhorse for the analysis of monetary policy, referred to as New Keynesian.

Meantime, one strand of the economic analyzed unemployment as search and matching process between households and firms. Early contributions to the current vintage of search and matching models include [Diamond \(1982a\)](#), [Diamond \(1982b\)](#), [Mortensen \(1982\)](#) [Pissarides \(1984\)](#). These models offers a central role to the interactions between firms and households in terms of job matching and wage negotiating. However, these contributions were partial equilibrium, and were not yet embedded into a dynamic macroeconomic model. These literature is referred to as the Diamond-Mortensen-Pissarides models.

Since the 2000, economists started to include the DMP model into real business cycles models and new keynesian models as [Shimer \(2005\)](#) and [Hall \(2005\)](#). This note aims at presenting the analytics of such model with a low set of frictions.

Since then, there have been a large number of estimated version of the labor matching model. See [Gertler and Trigari \(2009\)](#), [Galí et al. \(2012\)](#) and [Christiano et al. \(2016\)](#).

2. The model

2.1 Households

Suppose a welfare maximizing household enjoying current and future consumption and dislike working with respective curvatures $\sigma_C, \sigma_H > 0$. The j -th household maximizes the following stream of utilities discounted at a fixed rate β :

$$\max_{\{c_{jt}, b_{jt}, n_{jt}\}} \sum_{\tau=0}^{\infty} \beta^{\tau} E_t \left\{ \frac{\varepsilon_{t+\tau}^C c_{jt+\tau}^{1-\sigma_C}}{1-\sigma_C} - \chi n_{jt+\tau} \right\} \quad (1)$$

where c_{jt} is consumption, n_{jt} is the fraction of employed workers within the family j , and $\chi > 0$ is a shift parameter allowing to pin down the amount the employment rate in the deterministic steady state. The AR(1) shock ε_t^C captures exogenous changes in the utility obtained from consumption. Assuming that the size of the working age population is one, the number of family number is also one, so the shared unemployed within the family is given by:

$$n_{jt} = 1 - u_{jt} \quad (2)$$

The law of motion of employment is given by:

$$n_{jt} = (1 - \rho) n_{jt-1} + x_t u_{t-1} \quad (3)$$

where $\rho \in [0, 1]$ is the separation rate with the term ρn_{jt-1} that can be interpreted as the frictional unemployment. The term $x_t u_{t-1}$ is the number of unemployed workers who have found a job between $t-1$ and t , while $(1 - x_t) u_{t-1}$ is the fraction who stays unemployed between $t-1$ and t . Here, the hiring rate x_t is decided by firms, not by the household this is why there is no j subscript. Similarly, household j is so small that its own employment decisions has no implication on the unemployment rate u_{t-1} .

The budget constraint in real terms reads as:

$$c_{jt} + i_{jt} + f(\Delta i_{jt}) i_{jt-1} + b_{jt} = r_{t-1} b_{jt-1} + n_{jt} w_t + z_t k_{jt-1} + (1 - n_{jt}) \bar{b} - T_{jt}, \quad (4)$$

where b_{jt} is the amount of real bonds, r_{t-1} is the pre-determined real interest rate of real government bonds b_{jt-1} , w_t the real wage is computed as $w_t = W_t/P_t$ and a lump-sum tax financing the government T_{jt} . Here, the fraction of the family in employment is remunerated by the real wage w_t while the fraction of unemployed $1 - n_{jt}$ obtains an unemployment insurance $\bar{b}$. Note that i_{jt} is the flow of investment subject to adjustment costs with $f(x) = 0.5\kappa(x-1)^2$ and $\Delta i_{jt} = i_{jt}/i_{jt-1}$. Physical capital reads as follows:

$$\varepsilon_t^i i_{jt} = k_{jt} - (1 - \delta) k_{jt-1}. \quad (5)$$

Because the number of households is one, we drop the subscript j . The Lagrangian problem reads as:

$$\begin{aligned} \mathcal{L}_t = & \sum_{\tau=0}^{\infty} \beta^{\tau} \left[\varepsilon_{t+\tau}^C \frac{C_{t+\tau}^{1-\sigma_C}}{1-\sigma_C} - \chi n_{t+\tau} \right] \\ & + \beta^{\tau} \lambda_{t+\tau}^C \left[r_{t-1+\tau} b_{t-1+\tau} + n_{t+\tau} w_{t+\tau} + (1 - n_{t+\tau}) \bar{b} + z_{t+\tau} k_{t-1+\tau} - c_{t+\tau} - f(\Delta i_{t+\tau}) i_{t-1+\tau} - i_{t+\tau} - b_{t+\tau} - T_{t+\tau} \right] \\ & + \beta^{\tau} \lambda_{t+\tau}^C V_{t+\tau}^H \left[(1 - \rho) n_{t-1+\tau} + x_{t+\tau} u_{t-1+\tau} - n_{t+\tau} \right] \\ & + \beta^{\tau} \lambda_{t+\tau}^C q_{t+\tau} \left[k_{t+\tau} - (1 - \delta) k_{t-1+\tau} - \varepsilon_{t+\tau}^i i_{t+\tau} \right] \end{aligned}$$

where λ_t^C , q_t and V_t^H are lagrangian multipliers on each constraints. Note that both V_t^H and q_t are pre-multiplied by the marginal utility of consumption to express unemployment in terms of current utility-equivalent consumption.

First order conditions are given by:

$$c_t : \varepsilon_t^C c_t^{-\sigma_C} - \lambda_t^C = 0 \quad (6)$$

$$b_t : \beta E_t \{ \lambda_{t+1}^C \} r_t - \lambda_t^C = 0 \quad (7)$$

$$n_t : -\chi + \lambda_t^C (w_t - \bar{b}) - \lambda_t^C V_t^H + \beta \lambda_{t+1}^C (1 - \rho) V_{t+1}^H = 0 \quad (8)$$

$$i_t : -\lambda_t^C (1 + f'(\Delta i_t)) - \lambda_t^C q_t \varepsilon_t^i - \beta \lambda_{t+1}^C f'(\Delta i_{t+1}) = 0 \quad (9)$$

$$k_t : \beta \lambda_{t+1}^C z_{t+1} + \lambda_t^C q_t - \beta \lambda_{t+1}^C q_{t+1} (1 - \delta) = 0 \quad (10)$$

2.2 Firms

Firms technology is given by the following production function

$$y_{it} = \varepsilon_t^A k_{it-1}^\alpha n_{it}^{1-\alpha}, \quad (11)$$

where ε_t^A is the a technology shock, k_{t-1} is the physical stock of capital with fixed intensity of $\alpha \in [0; 1]$ and n_t is the employment demand.

As for households, the law of motion of employment:

$$n_{it} = (1 - \rho) n_{it-1} + x_{it} u_t, \quad (12)$$

where x_{it} is the hiring rate of the i-th firm that premultiply the stock of unemployed.

Profits reads as:

$$\Pi_t = y_{it} - w_t n_{it} - z_t k_{it-1} - G(x_{it}) u_t \quad (13)$$

where the cost of increasing hirings reads as in [Galí \(2010\)](#) and [Blanchard and Gali \(2010\)](#):

$$G(x_{it}) = \frac{\Gamma}{1 + \phi} x_{it}^{1+\phi}$$

where ϕ is the elasticity of hiring costs with respect to the hiring rate, and Γ is a constant which allows to pin down the total cost of hiring to one percent of total production as in [Blanchard and Gali \(2010\)](#). Variable x_{it} is the probability of hiring a new worker, the associated costs can be interpreted as monitoring cost paid by firms to find a potential new worker.

The dynamic Lagrangian problem reads as:

$$\begin{aligned} \mathcal{L}_t = & E_t \sum_{\tau=0}^{\infty} \beta^s \frac{\lambda_{t+\tau}^C}{\lambda_t^C} \{ [y_{it+\tau} - w_{t+\tau} n_{it+\tau} - z_{t+\tau} k_{it-1+\tau} - G(x_{it+\tau}) u_{t+\tau}] \\ & + V_{t+\tau}^P [(1 - \rho) n_{it-1+\tau} + x_{it+\tau} u_{t+\tau} - n_{it+\tau}] \} \end{aligned}$$

Note that $\beta^s \lambda_{t+\tau}^C / \lambda_t^C$ is the stochastic discount factor. As households are shareholders of firms because they own the physical capital, firms has the same discounting scheme as households. Also note that the firm takes as given the labor market situation (here measured as the unemployment rate) u_t , this is because each firm is so small that it is assumed that they cannot influence individually the aggregate unemployment rate.

$$\begin{aligned}
 n_{it} : (1 - \alpha) \frac{y_{it}}{n_{it}} - w_{it} - V_t^P + \beta \frac{\lambda_{t+1}^C}{\lambda_t^C} (1 - \rho) V_{t+1}^P &= 0 \\
 x_{it} : -G'(x_{it}) u_t + V_t^P u_t &= 0 \\
 k_{it-1} : \alpha \frac{y_{it}}{k_{it-1}} - z_t &= 0
 \end{aligned}$$

The standard hiring decision rate is given by this standard no arbitrage condition:

$$\underbrace{G'(x_{it})}_{\text{Marginal cost of hiring}} = \underbrace{(1 - \alpha) \frac{y_{it}}{n_{it}} - w_{it}}_{\text{marginal profit of hiring}} + \underbrace{\beta \frac{\lambda_{t+1}^C}{\lambda_t^C} (1 - \rho) G'(x_{it+1})}_{\text{Future gains from keeping the worker employed}}$$

2.3 Wage bargaining

Bilateral bargaining between firms and workers. Following form:

$$S_t = \underbrace{(V_t^H)}_{\text{household marginal utility}}^\eta \underbrace{(V_t^P)}_{\text{firm marginal profit}}^{1-\eta}$$

where η is the relative bargaining power of households. Period-by-period Nash bargaining implies that the firm and each of its workers determine the wage in period t by solving the problem:

$$\max_{\{w_t\}} S_t \tag{14}$$

The solution to that problem implies the following constant share rule:

$$\begin{aligned}
 \frac{\partial S_t}{\partial w_t} = 0 &\Leftrightarrow \eta \frac{\partial V_t^H}{\partial w_t} \frac{S_t}{V_t^H} + (1 - \eta) \frac{\partial V_t^P}{\partial w_t} \frac{S_t}{V_t^P} = 0 \\
 &\Leftrightarrow \eta V_t^P = (1 - \eta) V_t^H
 \end{aligned}$$

In order to take into account some exogenous change in the negotiation power of households, we augment η by an AR(1) shock, denoted ε_t^m , as follows:

$$\varepsilon_t^m \eta V_t^P = (1 - \varepsilon_t^m \eta) V_t^H$$

To illustrate how this equations determine the wage, we can rewrite substitute V_t^P and V_t^H :

$$w_t = \eta \left[(1 - \alpha) \frac{y_{it}}{n_{it}} + m_{t,t+1} V_{t+1}^P \right] + (1 - \eta) \left[-\chi / \lambda_t^C - b + m_{t,t+1} V_{t+1}^H \right]$$

where $m_{t,t+1} = \beta \lambda_{t+1}^C / \lambda_t^C (1 - \rho)$. It is straightforward to see that the wage is the result between the desired wage for firms and household weighted by the bargaining power of each agent.

2.4 Authorities

Public spending G_t is exogenous, as it follows an AR(1) shock process ε_t^G , which is expressed as a constant portion of output:

$$G_t = g^y \bar{Y} \varepsilon_t^G.$$

The balance sheet reads as:

$$r_{t-1} b_{t-1} + G_t + b u_t = b_t + T_t,$$

where b_t are government bonds.

2.5 Equilibrium conditions

And total demand is given by:

$$Y_t = C_t + I_t + G_t + G(x_t) u_t. \quad (15)$$

Unemployment rate:

$$u_t = 1 - n_t \quad (16)$$

Note that as in Blanchard & Gali (2012), it is assumed that hiring costs are set to match 1% of output in the deterministic steady state such as $G(\bar{x}) \bar{u} = 0.01 \bar{Y}$.

References

- Blanchard, O., Gali, J., 2010. Labor markets and monetary policy: A new keynesian model with unemployment. *American economic journal: macroeconomics* 2 (2), 1–30.
- Christiano, L. J., Eichenbaum, M. S., Trabandt, M., 2016. Unemployment and business cycles. *Econometrica* 84 (4), 1523–1569.
- Diamond, P. A., 1982a. Aggregate demand management in search equilibrium. *Journal of political Economy* 90 (5), 881–894.
- Diamond, P. A., 1982b. Wage determination and efficiency in search equilibrium. *The Review of Economic Studies* 49 (2), 217–227.
- Galí, J., 2010. Monetary policy and unemployment. In: *Handbook of Monetary Economics*. Vol. 3. Elsevier, pp. 487–546.
- Galí, J., Smets, F., Wouters, R., 2012. Unemployment in an estimated new keynesian model. *NBER macroeconomics annual* 26 (1), 329–360.
- Gertler, M., Trigari, A., 2009. Unemployment fluctuations with staggered nash wage bargaining. *Journal of political Economy* 117 (1), 38–86.
- Hall, R. E., 2005. Employment fluctuations with equilibrium wage stickiness. *American economic review* 95 (1), 50–65.
- Keynes, J. M., 1937. The general theory of employment. *The quarterly journal of economics* 51 (2), 209–223.
- Lucas, R., 1976. Econometric policy evaluation: A critique. the phillips curve and labour markets. k. brunner and ah meltzer.
- Mortensen, D. T., 1982. The matching process as a noncooperative bargaining game. In: *The economics of information and uncertainty*. University of Chicago Press, pp. 233–258.
- Phelps, E. S., 1967. Phillips curves, expectations of inflation and optimal unemployment over time. *Economica*, 254–281.
- Pissarides, C. A., 1984. Search intensity, job advertising, and efficiency. *Journal of labor Economics* 2 (1), 128–143.
- Shimer, R., 2005. The cyclical behavior of equilibrium unemployment and vacancies. *American economic review* 95 (1), 25–49.